



Software Requirements Specifications

**Online Fact-Checking System
(Sebenarnya.my)**

Version: 1.0.0

Date: 4 January 2024

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my 
Date: 4 January 2026	Test Design Specification ID: SeBenarnya_SRS_1.0.0	

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Version History

Version	Release Date	Section	Amendments
1.0	4 January 2026	All	Original Document

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1.0 Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the SeBenarnya.my System.

This system aims to automatically verify, classify, and analyze online news articles to identify misinformation. It integrates Artificial Intelligence (AI), Natural Language Processing (NLP), and user analytics to detect patterns of false information and enhance media credibility.

1.2 Scope

The SeBenarnya.my System is a web-based platform designed to classify submitted news, detect multiple languages, and monitor user behavior.

It supports fact-checkers, administrators, and the public in ensuring that published content is credible, verified, and traceable.

1.3 Definition & Acronyms

Definition

Admin An admin is the system operator responsible for managing news verification data, monitoring flagged users, and reviewing classification results generated by the system.

User A user is a member of the public who submits news articles to the

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SeBenarnya.my platform for verification and views the classification results.

Fact-Checker

A fact-checker is a verified person or organization responsible for confirming or correcting the AI-generated classification results.

News Submission

A news submission refers to the content (article, post, or text) uploaded by a user for verification by the system.

Classification Result

The result produced by the AI-based module indicating whether the news is True, False, or Unverified.

Credibility Score

A numerical score assigned to users or sources to represent their trustworthiness, calculated based on submission history and verification results.

Language Classifier

A subsystem that detects and processes news content according to its language (example: Malay, English, Mandarin).

Acronyms

AI	Artificial Intelligence
NLP	Natural Language Processing
SRS	Software Requirement Specification
UI	User Interface

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1.4 References

This document is prepared in reference to the following documents:

1. ATMS Software Requirement Specifications. (2013, April 4). Malaysian Software Testing Board.
2. Lane, C., & Krüger, N. (2021, December 16). *How to Write a Software Requirements Specification (SRS Document)*. Retrieved 28 December 2022, from [How to Write an SRS \(Software Requirements Specification Document\) \(perforce.com\)](#)

1.5 Overview

This document consists of three sections:

1. Section 1 - Describes the purpose, scope, definitions, acronyms, and references.
2. Section 2 - Provides an overall description of the system, including interfaces, product features, and user characteristics.
3. Section 3 - Details the specific functional and non-functional requirements to be implemented in the MySeBenarnya System.

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2.0 Overall Description

2.1 Product Perspective

2.1.1 User Interfaces

Requirement ID	Description	Priority	Author
REQ-UI-001	The submission page shall allow users to input news content and upload optional URLs or images for classification.	High	Nur Izzah Dania
REQ-UI-002	The system shall provide multi-language support for user interfaces and generated reports.	High	Nur Izzah Dania
REQ-UI-003	The admin dashboard shall display all verified news records with filters for date, source, and classification type.	High	Meor Muhammad Anuar
REQ-UI-004	The fact-checker dashboard shall allow reviewers to approve or reject AI classifications.	High	Nur Liyana Athirah

2.1.2 Hardware Interfaces

Hardware Interface	Description
Device	Laptop
Processor	AMD Ryzen™ 5 7520U Mobile Processor
System Type	64-bit operating system, x64-based processor
Device Memory	8GB

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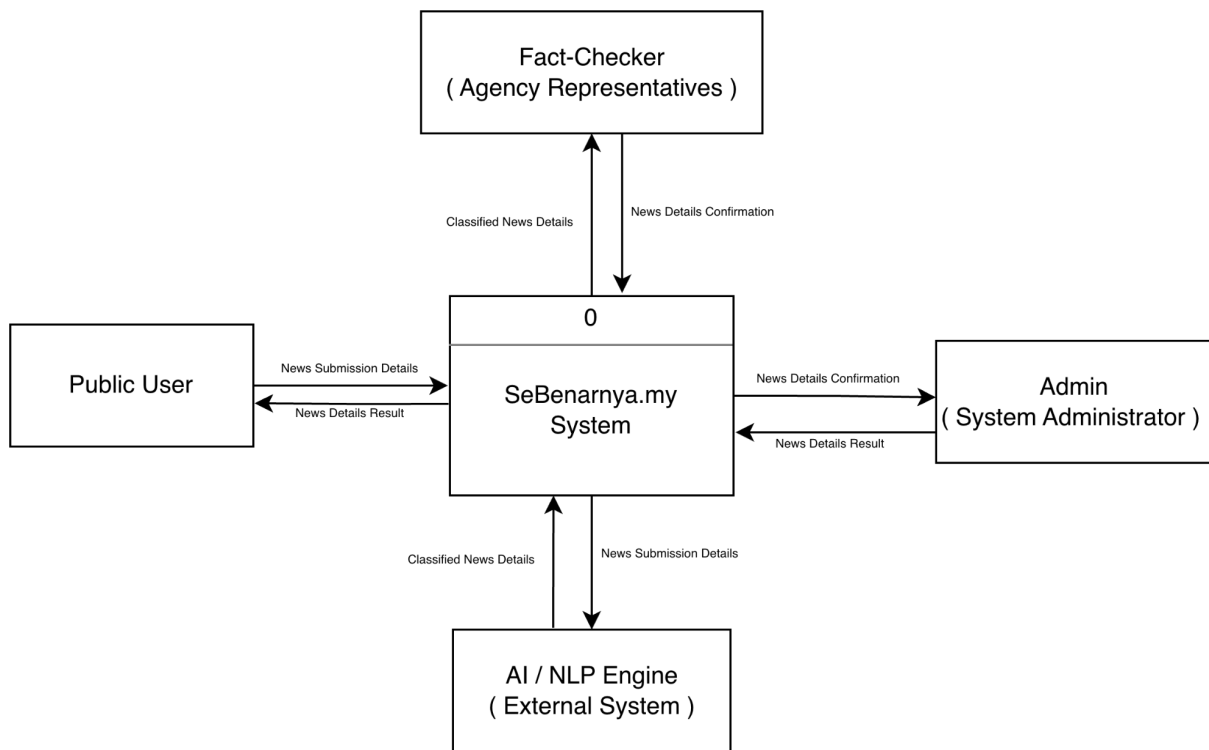
2.1.3 Communication Interfaces

Communication Interface	Description
HTTPS	Secure communication protocol
REST API	Interface for AI engines and backend processing
MySQL	Database communication

2.2 Product Feature

Function ID	Feature	Description	Accessible By
F001	Multi-Language Detection	Detects the language of submitted news content (Malay, English, Mandarin) before classification.	System
F002	Language Routing	Routes detected language content to the appropriate language-specific classifier.	System
F003	Cross-Language Duplicate Detection	Detects duplicate or semantically similar news content across different languages.	System
F004	Translation Validation	Validates the accuracy and quality of translated content before analysis.	System
F005	AI-Based News Classification	Automatically classifies news content as True, False, Misleading, or Unverified.	System
F006	Multi-Language Reporting	Generates classification reports in the user's selected interface language.	System
F007	User submission	Allows users to submit news content for verification.	User
F008	Fact-Check verification workflow	Allows fact-checkers to approve or correct AI-generated classification results.	Fact-Checker

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2.3 User Characteristics

There are mainly three potential roles targeted by SeBenarnya.my which are Admin, User and Fast-Checker. Below is the table showing the characteristics and the expected level of knowledge for each role.

Role	Description	Requirements
Admin	System manager who oversees submissions, user activity, and official reports.	Basic technical & management skills
User	Public individual who checks the authenticity of news.	Able to use the website and basic submission actions
Fact-Checker	Verified agency personnel reviewing accuracy.	Understanding of news analysis and verification

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3.0 Specific Requirements

3.1 External Interfaces

To be defined.

3.2 Functional Requirement

This section details the functional requirements of the SeBenarnya.my system, starting with the overall requirement, followed by the requirements of each feature of the system.

3.2.1 AI-Based News Classification

Requirement ID	REQ-01-001	Version	1.0
Description	The system shall classify submitted news as True, False, Misleading, or Unverified using an AI model.		
Author	Nur Liyana Athirah Binti Zailani		

Requirement ID	REQ-01-002	Version	1.0
Description	The system shall automatically retrain the AI model using newly verified fact-checker results.		
Author	Nur Liyana Athirah Binti Zailani		

Requirement ID	REQ-01-003	Version	1.0
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Description	The system shall detect rephrased or repeated patterns of previously known fake news.
Author	Nur Liyana Athirah Binti Zailani

Requirement ID	REQ-01-004	Version	1.0
Description	The system shall generate an explainable AI output showing reasoning for its classification.		
Author	Nur Liyana Athirah Binti Zailani		

Requirement ID	REQ-01-005	Version	1.0
Description	The system shall display an AI confidence score (0–100%) to Fact-Checkers and Admins only, while Public Users shall only see the verified classification result.		
Author	Nur Liyana Athirah Binti Zailani		

Requirement ID	REQ-01-006	Version	1.0
Description	The system shall validate AI-based news classification results by ensuring that explainable AI output is relevant, consistent with the input content, and free from hallucinated or unrelated explanations. The system shall also handle AI service failures, low-confidence classifications, and inconsistent outputs gracefully by flagging results for manual review and logging incidents.		

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Author	Nur Liyana Athirah Binti Zailani
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Requirement ID	REQ-01-007	Version	1.0
Description	The system shall ensure that AI-based news classification does not classify adversarially manipulated or ambiguous content as “True” and shall flag such content as “Misleading”, “Unverified”, or for manual review.		
Author	Nur Liyana Athirah Binti Zailani		

3.2.2 Multi-Language Detection & Verification

Requirement ID	REQ-02-001	Version	1.0
Description	The system shall automatically detect the language of each submitted news content (Malay, English, or Mandarin) with high accuracy to ensure proper downstream processing.		
Author	Nur Izzah Dania Binti Hipni Shahlizal		

Requirement ID	REQ-02-002	Version	1.0
Description	The system shall route every submission to its respective language-specific classifier module to perform more precise categorization and analysis.		
Author	Nur Izzah Dania Binti Hipni Shahlizal		

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Requirement ID	REQ-02-003	Version	1.0
Description	The system shall detect duplicate or highly similar news content submitted across different languages by comparing cross-language semantic meaning.		
Author	Nur Izzah Dania Binti Hipni Shahlizal		

Requirement ID	REQ-02-004	Version	1.0
Description	The system shall validate the accuracy and quality of translated content (when translation is required) before performing any classification or verification processes.		
Author	Nur Izzah Dania Binti Hipni Shahlizal		

Requirement ID	REQ-02-005	Version	1.0
Description	The system shall generate multi-language classification reports based on the user's preferred interface language to ensure clarity and accessibility.		
Author	Nur Izzah Dania Binti Hipni Shahlizal		

3.2.3 User Behavior Analytics & Reputation System

Requirement ID	REQ-03-001	Version	1.0
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Description	The system shall track and store all user submissions linked to their account profile.
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh

Requirement ID	REQ-03-002	Version	1.0
Description	The system shall automatically count the number of confirmed false submissions per user..		
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh		

Requirement ID	REQ-03-003	Version	1.0
Description	The system shall issue warnings or temporary suspension for users who repeatedly submit false or harmful content.		
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh		

Requirement ID	REQ-03-004	Version	1.0
Description	The system shall maintain a dynamic reputation score that increases with correct submissions and decreases with verified false submissions.		

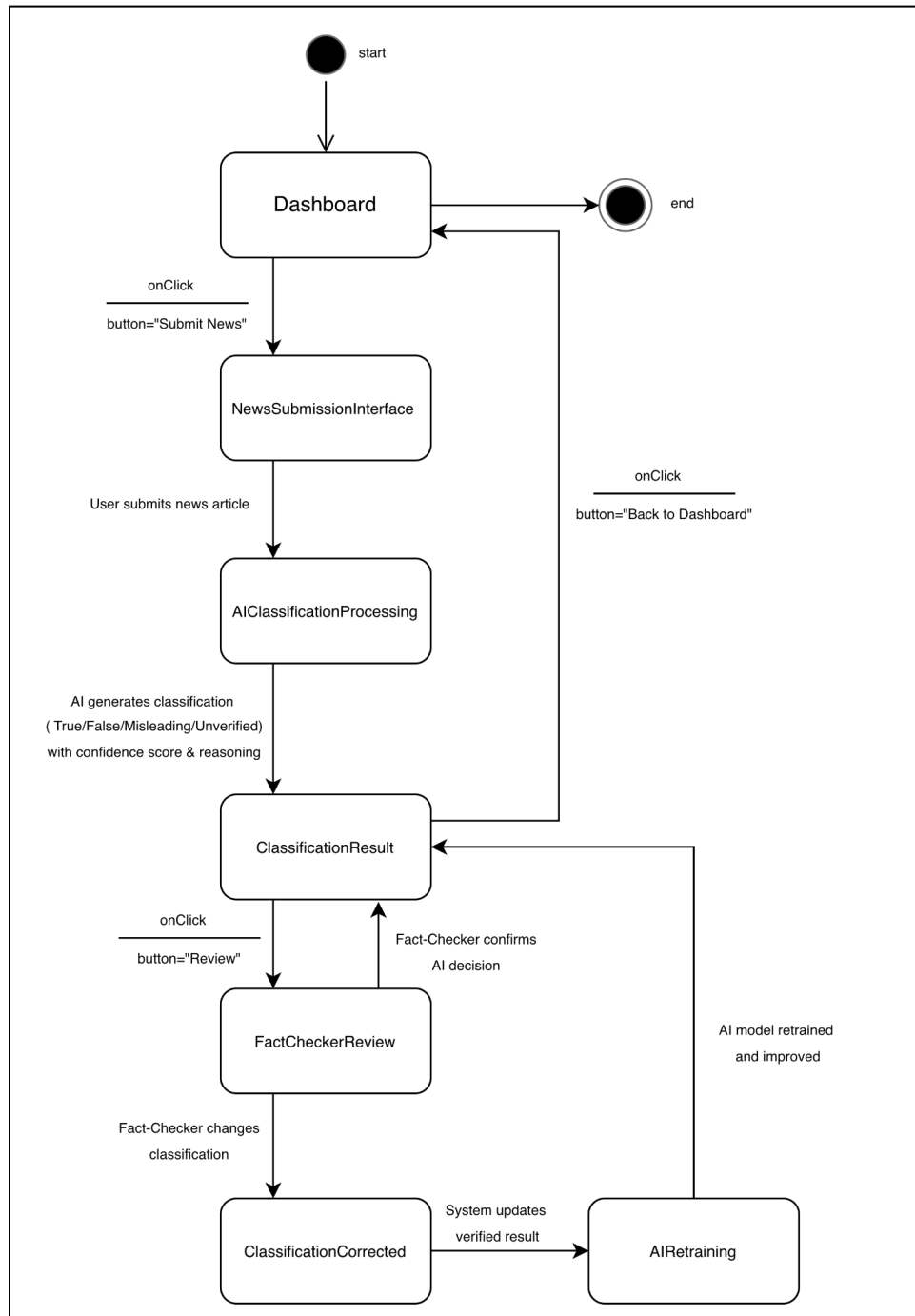
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Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh
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Requirement ID	REQ-03-005	Version	1.0
Description	The system shall send automated alerts (email/app notifications) when a user approaches violation thresholds.		
Author	Meor Muhammad Anuar Firdaus bin Ainol Salleh		

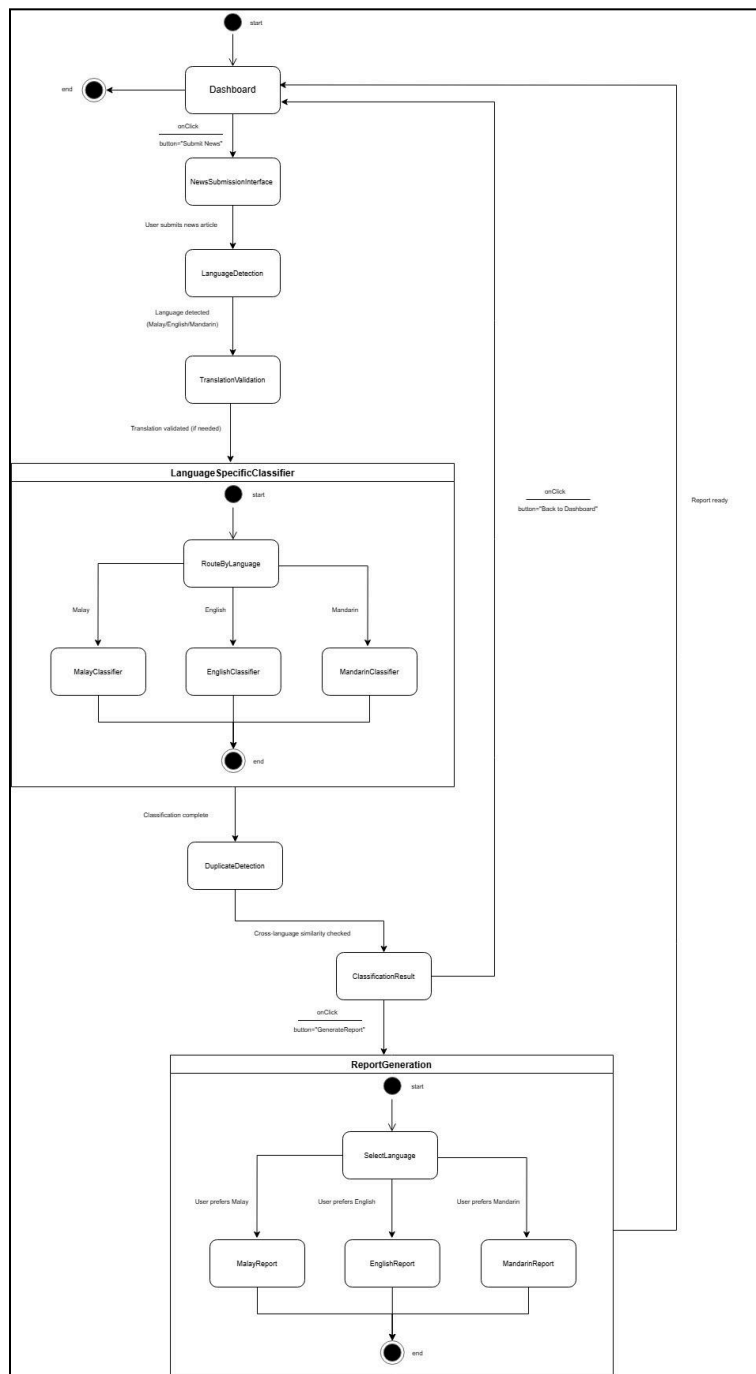
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3.2.1 AI-Based News Classification [REQ-01-001]



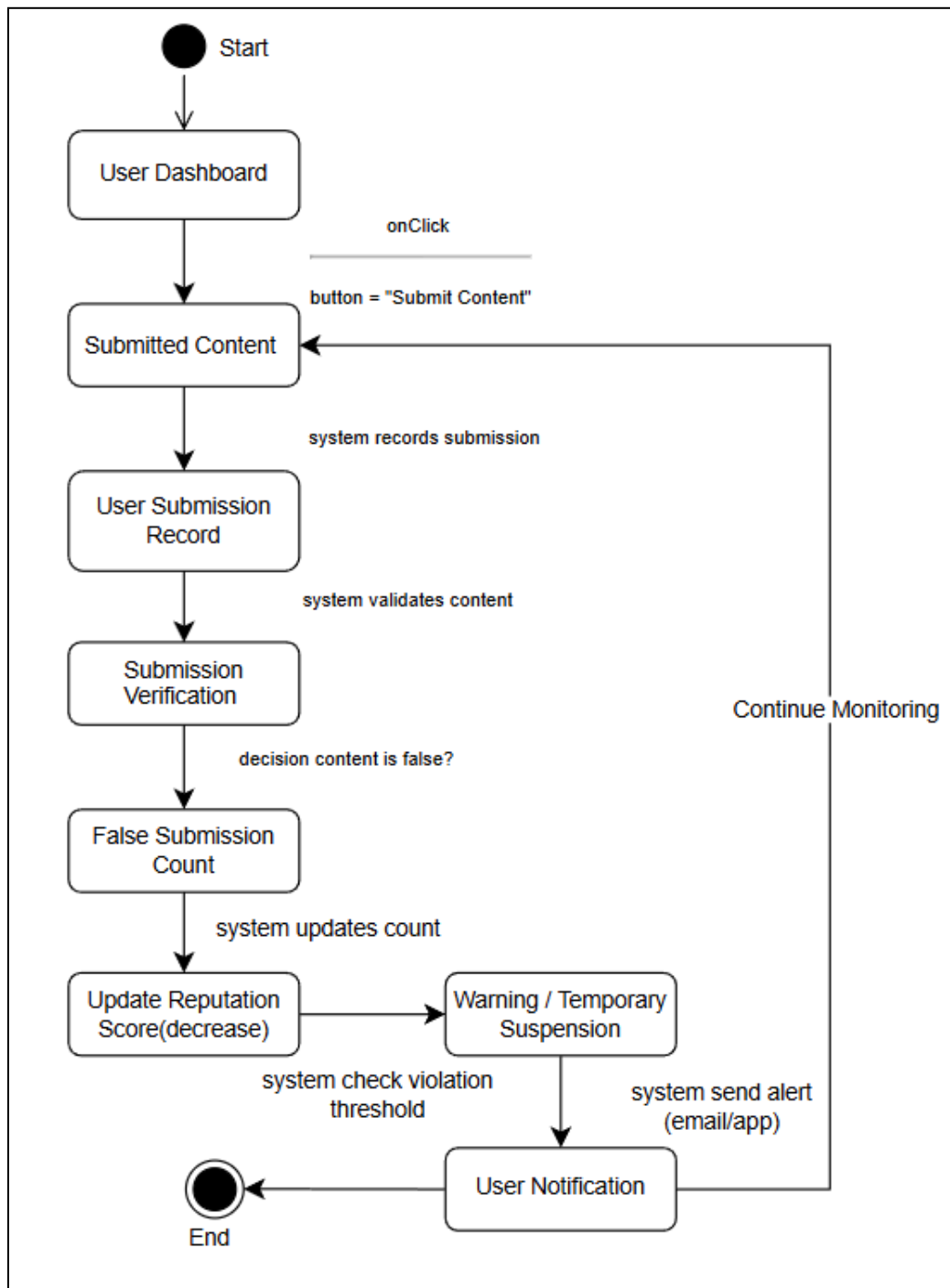
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3.2.2 Multi-Language Detection & Verification [REQ-02-001]



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3.2.3 User Behavior Analytics & Reputation System [REQ-03-001]



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3.3 Design Constraints

1. User Interface Constraints

- a. The user interface of the MySebenarnya system shall have a responsive design that adapts to different screen sizes (desktop, tablet, mobile).
- b. The system shall maintain intuitive navigation to ensure ease of use for public users, fact-checkers, and administrators.

2. Security Constraints

- a. All user accounts (Admin, Fact-Checker, User) must register secure passwords containing a combination of alphabets, numbers, and symbols.
- b. Only authorized users with valid credentials shall access the system modules.

3. Database Constraints

- a. The database must be scalable and capable of handling increasing submissions and classification records over time.

4. Compatibility Constraints

- a. The system must be compatible with major browsers such as Chrome, Safari, Edge, Firefox, and Opera.

5. Performance Constraints

- a. The system shall process core functionalities such as news submission, classification retrieval, and dashboard loading within acceptable time (≤ 3 seconds).

6. Communication Interface Constraints

- a. The system shall use HTTPS as the communication protocol.

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- b. The system shall use REST API standards for communication between AI engines and backend services.

3.4 Software System Attributes

3.4.1 Accuracy

Requirement ID	Description	Priority	Author
REQ-QA001	The AI model shall accurately classify submitted news as True, False, Misleading, or Unverified.	High	Nur Liyana Athirah Binti Zailani
REQ-QA002	The system shall ensure accurate retraining of the AI model using newly verified fact-checker results.	High	Nur Liyana Athirah Binti Zailani
REQ-QA003	The AI module shall accurately detect rephrased or repeated patterns of previously known fake news.	High	Nur Liyana Athirah Binti Zailani
REQ-QA004	The system shall generate accurate explainable AI outputs that reflect the reasons behind each classification.	High	Nur Liyana Athirah Binti Zailani
REQ-QA005	The AI confidence score (0–100%) shall accurately represent the system’s certainty level for Fact-Checkers and Admins.	High	Nur Liyana Athirah Binti Zailani

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REQ-QA006	The system shall accurately detect the primary language (Malay, English, or Mandarin) of every news submission with high reliability.	High	Nur Izzah Dania Binti Hipni Shahlizal
REQ-QA007	The system shall accurately route detected languages to the correct language-specific classification model with minimal language-routing errors.	High	Nur Izzah Dania Binti Hipni Shahlizal
REQ-QA008	The system shall accurately identify cross-language duplicates or semantically similar news content across Malay, English, and Mandarin submissions.	High	Nur Izzah Dania Binti Hipni Shahlizal
REQ-QA009	The translation validation feature shall ensure that meaning, intent, and factual context are preserved accurately before the AI performs analysis.	High	Nur Izzah Dania Binti Hipni Shahlizal
REQ-QA010	The system shall generate accurate multi-language classification reports that maintain terminology precision based on the user's selected interface language.	High	Nur Izzah Dania Binti Hipni Shahlizal
REQ-QA011	The system shall accurately link each user submission to the correct user account profile without mismatch.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh

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REQ-QA012	The system shall accurately count and record only confirmed false submissions for each user.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QA013	The system shall accurately calculate and update the user reputation score based on verified correct and false submissions.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QA014	The system shall accurately determine when a user reaches predefined violation thresholds.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QA015	The system shall accurately trigger warnings, suspensions, and alerts only for users who meet violation criteria.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh

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3.4.2 Reliability

Requirement ID	Description	Priority	Author
REQ-QR001	The AI module shall reliably generate classification results for all submitted news without system failure.	High	Nur Liyana Athirah Binti Zailani
REQ-QR002	The AI model shall consistently detect previously known misinformation patterns across repeated or rephrased inputs.	High	Nur Liyana Athirah Binti Zailani
REQ-QR003	The AI explainability feature shall reliably produce reasoning outputs for every classification generated.	High	Nur Liyana Athirah Binti Zailani
REQ-QR004	The AI model shall maintain reliable performance after retraining, ensuring classification consistency over time.	High	Nur Liyana Athirah Binti Zailani
REQ-QR005	The system shall reliably store and deliver accurate confidence scores to Fact-Checkers and Admins during every classification request.	High	Nur Liyana Athirah Binti Zailani
REQ-QR006	The Multilanguage AI module shall reliably process and classify news in all supported languages without system failure.	High	Nur Izzah Dania binti Hipni Shahlizal
REQ-QR007	The AI model shall consistently detect misinformation patterns across different languages, including paraphrased or restructured inputs.	High	Nur Izzah Dania binti Hipni Shahlizal
REQ-QR008	The translation and classification pipeline shall reliably output results even under high user load or rapid repeated requests.	High	Nur Izzah Dania binti Hipni Shahlizal

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REQ-QR009	The AI reasoning/explanation feature shall reliably generate justification outputs for every classification across all supported languages.	High	Nur Izzah Dania binti Hipni Shahlizal
REQ-QR010	The system shall reliably store and return multilingual confidence scores to Fact-Checkers and Admins for every classification request.	High	Nur Izzah Dania binti Hipni Shahlizal
REQ-QR011	The system shall reliably store all user submission records without data loss or corruption.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QR012	The system shall consistently track false submission counts across multiple user activities and sessions.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QR013	The reputation scoring mechanism shall reliably maintain consistent score updates over time.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QR014	The system shall reliably enforce warnings or temporary suspensions when violation thresholds are reached.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh
REQ-QR015	The notification service shall reliably send alerts to users when violation thresholds are approached or exceeded.	High	Meor Muhammad Anuar Firdaus bin Ainol Salleh